Raj Dubey

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EDUCATION

Pune University Pune, Maharashtra

Bachelor of Engineering in Information Technology (8.86 CGPA) *Aug 2019 – Aug 2023*

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy), SQL (Teradata Shell Scripting), JavaScript, HTML/CSS, Vue.js.

Backend: FastAPI, asynchronous programming, REST API development, API integrations, request validation, pagination, RBAC, error handling

Cloud and delivery: AWS, GCP, Docker, GitHub Actions, Argo CD

AI / ML & GenAI: RAG, Chroma DB, Ollama, Semantic Testing using LLMs, MCP servers.

EXPERIENCE

Teradata, Magarpatta, Pune: ***AI Engineer*** **Sep 2025 – Present**

- **Production AI Systems & Agentic Workflows: -** Engineered a production-grade AI system advisor integrating **Large Language Models (LLMs**) with observability pipelines (ELK Stack) to automate root-cause analysis, significantly reducing **Mean Time to Resolution (MTTR**). Designed agentic workflows to autonomously monitor system health and generate actionable infrastructure optimizations.
- Build agentic workflows to autonomously monitor system health, analyze infrastructure logs, and generate actionable optimization strategies for cloud environments.
- Connected the advisor to **MCP servers and custom APIs**, enabling structured tool use for secure, read-only Kubernetes diagnostics Supported production delivery through **GitOps-based Argo CD releases** and blue-green deployment.
- Applied access controls and **read-only tool boundaries** to keep diagnostic actions scoped and reduce the risk of unintended changes.

Backend Engineer | FastAPI and system design **Sep 2023-25**

- Built and **maintained asynchronous FastAPI REST** endpoints for BaaS jobs, job details, execution history, and scheduled runs, integrating with authenticated upstream APIs.
- Used **FastAPI dependency injection** for service access and **RBAC checks**, with typed **path/query parameters**, input constraints, pagination, Understand OAuth2/JWT, secret management, OWASP API security risks, tenant isolation, load testing, profiling, cache invalidation, capacity planning, and system-design trade-offs.
- **Designed a Redis-backed**, precomputed read path: a background worker refreshes site-job data so common API reads avoid repeatedly calling the upstream service.
- Improved refresh reliability with bounded **concurrency, request pacing, latency-aware backoff**, and a Redis lock to prevent overlapping full refreshes, added upstream timeouts and retries, **normalized API responses, and applied platform/job-type** filtering to cached results.
- Build database design, normalization, indexing, transactions, isolation levels, and schema migrations.
- Implemented API versioning, cursor pagination, **request-level rate limiting**, structured logging, metrics, distributed tracing, and Open Telemetry.

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PROJECTS

(LINKS TO PROJECT ARE HYPERLINKED TO THE PROJECT TITLE)

End-to-End RAG API Pipeline | Python, FastAPI, Docker, Kubernetes, GitHub Actions

- Designed and implemented a **production-ready Retrieval-Augmented Generation (RAG) API** using **FastAPI**, integrating **ChromaDB** for vector search and **Ollama** for local LLM inference
- Deployed and orchestrated the application on a **local Kubernetes cluster (Minikube)** using **Deployments and Services** for scalability, self-healing, and stable networking.
- Automated embedding rebuilds and semantic validation using GitHub Actions.
- Enforced **automated quality gates** to block regressions and prevent faulty deployments.
- Delivered a **fully automated MLOps workflow**, transforming a local prototype into a scalable, AI Systems.

MicroCloudPy | *Python, Fastapi, PostgreSQL, Helm, kubectl, MongoDB*

- Deploys a **Python-based microservices architecture** on **AWS Elastic Kubernetes Service (EKS)** using **Helm** and **kubectl**.
- Includes four core services — **auth-server, converter-module, Database-server** (PostgreSQL & MongoDB)
- Uses RabbitMQ for inter-service communication and message queuing.
- Demonstrates cloud-native deployment, scalability, and container orchestration with Kubernetes.
- Designed the architecture to be cloud-agnostic, allowing for seamless deployment on **AWS EKS** or **GCP GKE**.